J Neurol (2016) 263:1166–1177
DOI 10.1007/s00415-016-8119-y

ORIGINAL COMMUNICATION

Predictors of functional level and quality of life at 6 months after a first-ever stroke: the KOSCO study

Won Hyuk Chang[1] · Min Kyun Sohn[2] · Jongmin Lee[3] · Deog Young Kim[4] · Sam-Gyu Lee[5] · Yong-Il Shin[6] · Gyung-Jae Oh[7] · Yang-Soo Lee[8] · Min Cheol Joo[9] · Eun Young Han[10] · Chung Kang[1] · Yun-Hee Kim[1,11]

Received: 2 February 2016 / Revised: 4 April 2016 / Accepted: 4 April 2016 / Published online: 25 April 2016
© Springer-Verlag Berlin Heidelberg 2016

Abstract Predicting functional outcome and quality of life (QOL) is critical to the treatment of patients with stroke. The purpose of this study was to analyze the factors influencing functional status and QOL of stroke patients 6 months after a first-ever stroke. This study was an interim analysis of the Korean Stroke Cohort for Functioning and Rehabilitation Study, designed to include 10 years of follow-up for first-ever stroke patients. This study analyzed data from 2857 patients who completed face-to-face assessments with the functional independence measurement (FIM) and Euro Quality of Life-5D (EQ-5D) at 6 months after stroke onset. A multivariate regression analysis was used to analyze factors that potentially influenced FIM and EQ-5D results at 6 months after stroke. Of the patients in this study, 80.1 % suffered from ischemic stroke and 19.9 % experienced hemorrhagic stroke. The independent predictors of functional independency measured by FIM at 6 months after stroke were age, initial stroke severity, duration of hospitalization, and functional level at discharge in terms of motor, ambulation, and language. For QOL measured by EQ-5D at 6 months after stroke, age, duration of hospitalization, and motor function at discharge were significant predictors. In conclusion, proper treatment to achieve maximal functional gain at discharge may be an important factor in improving functional independency and QOL in chronic stage stroke survivors. These results provide useful information for establishing comprehensive and systematic care for stroke patients.

Keywords Stroke · Function · Independence · Quality of life · Prognosis · Cohort

✉ Yun-Hee Kim
yunkim@skku.edu; yun1225.kim@samsung.com

[1] Department of Physical and Rehabilitation Medicine, Center for Prevention and Rehabilitation, Heart Vascular Stroke Institute, Samsung Medical Center, Sungkyunkwan University School of Medicine, 50 Ilwon-dong, Seoul, Republic of Korea

[2] Department of Rehabilitation Medicine, School of Medicine, Chungnam National University, Daejeon, Republic of Korea

[3] Department of Rehabilitation Medicine, Konkuk University School of Medicine, Seoul, Republic of Korea

[4] Department and Research Institute of Rehabilitation Medicine, Yonsei University College of Medicine, Seoul, Republic of Korea

[5] Department of Physical and Rehabilitation Medicine, Chonnam National University Medical School, Gwangju, Republic of Korea

[6] Department of Rehabilitation Medicine, Pusan National University School of Medicine, Pusan National University Yangsan Hospital, Busan, Republic of Korea

[7] Department of Preventive Medicine, Wonkwang University, School of Medicine, Iksan, Republic of Korea

[8] Department of Rehabilitation Medicine, Kyungpook National University School of Medicine, Kyungpook National University Hospital, Daegu, Republic of Korea

[9] Department of Rehabilitation Medicine, Wonkwang University School of Medicine, Iksan, Republic of Korea

[10] Department of Rehabilitation Medicine, Jeju National University Hospital, Jeju National University School of Medicine, Jeju, Republic of Korea

[11] Department of Health Science and Technology, Department of Medical Device Management and Research, SAIHST, Sungkyunkwan University, Seoul, Republic of Korea

Introduction

Stroke is a major global cause of serious, long-term disability [1]. Many stroke survivors are left with some degree of stroke sequelae, and disability after stroke can be a significant burden on patients, caregivers, and society [2]. As the prevalence of stroke increases, the burden of disability is likely to become an increasingly important public health concern [3].

Predicting functional outcome measured with the level of activities of daily living (ADL) is critical to the treatment of patients with stroke [4]. Factors that accurately predict functional outcome after stroke are important for early stroke management in order to set suitable rehabilitation goals, enable early discharge planning, and correctly inform patients and relatives [5]. In the acute phase of stroke, the main predictors of outcome are stroke severity and patient age [6]. Additional important predictors include functional status prior to stroke onset and presence of comorbid medical conditions [7]. Stroke outcome is also affected by the development of stroke-related complications and extent of family and social support [8–10]. In addition, education years as an indicator of socioeconomic status can affect functional impairment after stroke [11–15]. One of the goals of stroke rehabilitation should be to improve the quality of life (QOL) in patients. Predictive factors are still not well understood for QOL in terms of the relationship between physical disability and QOL after stroke [16]. However, generalization of previous prognostic research is mostly limited because the predictive value of most determinants for outcome after stroke is indistinct [4]. To develop prediction models in patient subgroups and to improve the precision of validated models, large cohorts are needed. The Korean Stroke Cohort for Functioning and Rehabilitation (KOSCO) is a cohort of all acute first-ever stroke patients who were admitted to representative hospitals in nine distinct areas of Korea [17].

In this study, we analyzed this large cohort with subgroups of ischemic or hemorrhagic strokes and using a face-to-face functional assessment battery in order to develop a more accurate predictive model of outcome in stroke patients at discharge from the initial hospitalization. The goal of this study was to determine the predictive factors of functional level and QOL in the chronic phase in patients with a first-ever stroke.

Methods

The Korean Stroke Cohort for functioning and rehabilitation

The KOSCO study is a 10-year long-term follow-up study of stroke patients using a prospective multicenter design investigating residual disabilities, activity limitation, and QOL in patients who have suffered a first-ever stroke. While ischemic and hemorrhagic strokes are included, transient ischemic attacks are excluded. The detailed rationale and protocol of KOSCO was described in a previous article [17].

Written informed consent was obtained from all patients prior to inclusion in the study, and the study protocol was approved by the local ethics committee in each hospital.

To increase and maintain interrater reliability and accuracy, all raters in the KOSCO study underwent standardized training program at the beginning of and every 3 months during the course of the study. The initial training program with the 1-day workshop was performed three times before the enrollment. After the initial training program, only rater who passed the standardized examination for functional assessments including the on-line testing for FIM and K-MBI could participate in the data collection. The training program consisted of a 1-day workshop in addition to one-to-one education by an experienced rater for 1 week. The same regulation applied to an additional rater who participated in the data collection.

Selection of KOSCO participants

This study presents interim KOSCO study results, focusing on functional level and QOL of first-ever stroke patients 6 months after onset. Patients who were recruited to the KOSCO study group from August 2012 to October 2014 were analyzed in this study.

Functional outcome and quality of life at 6 months after stroke

In this study, functional outcome at 6 months after stroke was defined using the functional independence measure (FIM) [18]. The FIM assesses limitations in daily living activities, with scores ranging from 18 (completely dependent) to 126 (completely independent). FIM scores were grouped into six categories: group 1, total assistance (FIM, 18–39); group 2, maximal assistance (FIM, 40–59); group 3, moderate assistance (FIM, 60–79); group 4, minimal assistance (FIM, 80–99); group 5, supervision (FIM, 100–119); and group 6, independence (FIM, 120–126). In addition, the Euro Quality of Life (EQ)-5D [19, 20] was analyzed to assess QOL. EQ-5D scores were grouped into five categories: group 1, not at all satisfied (EQ-5D, less than 0.000); group 2, severe dissatisfied (EQ-5D, 0.000–0.3000); group 3, moderate dissatisfied (EQ-5D, 0.3000–0.6000); group 4, little dissatisfied (EQ-5D, 0.6000–0.9000); and group 5, satisfied (EQ-5D, more than 0.9000).

Baseline stroke severity and functional assessment at hospitalization discharge

Initial stroke severity was recorded using medical chart review at the time of hospital arrival based on the National Institute of Health Stroke Scale (NIHSS) [21] and Glasgow Coma Scale (GCS) [22] for ischemic and hemorrhagic strokes, respectively.

At discharge from hospitalization, the stroke severity of all patients was assessed using face-to-face methods. Functional assessments were carried out using the face-to-face functional assessment battery including the Korean Mini-Mental State Examination (K-MMSE) [23], Fugl-Meyer assessment (FMA) [24], functional ambulatory category (FAC) [25], American Speech-Language-Hearing Association National Outcome Measurement System Swallowing Scale (ASHA-NOMS) [26], and the Short Korean version of the Frenchay Aphasia Screening Test (Short K-FAST) [27].

Demographic and clinical characteristics of the patients at admission

Data on clinical characteristics such as comorbidity and data on demographic characteristics such as age group, sex, number of years of education, and prestroke dependency were documented by members of the hospital staff during acute treatment. Comorbidity was assessed using the combined condition- and age-related score in the Charlson comorbidity index (CCAS) [28]. Prestroke functional level was assessed using modified Rankin scale [29]. The following comorbidities were recorded during admission to the hospital: hypertension (systolic blood pressure >160 mmHg, diastolic blood pressure >90 mmHg, or history of hypertension or medical treatment); diabetes mellitus (elevated blood glucose level >126 mg/day or history of diabetes or medical treatment); coronary heart disease (documented by standard ECG or coronary imaging study or history of coronary heart disease or medical treatment); atrial fibrillation (documented by standard ECG, long-term ECG, or history of atrial fibrillation or medical treatment); and hyperlipidemia (elevated LDL cholesterol level >160 mg/dL, elevated total cholesterol level >240 mg/dL, or history of hyperlipidemia or medical treatment). Arrival time at the hospital and duration of first hospitalization were also recorded.

The course of the disease during admission was reported, including information on neurologic aggravation, complications during hospitalization, and hospitalization duration. Seven common medical complications after stroke were recorded during admission. In addition, the number of medical complications was calculated.

Statistical analysis

Descriptive statistics were used to provide information on patient demographic and clinical characteristics. Pearson correlation analysis was performed to assess the relationship between functional outcome and QOL at 6 months after stroke. The ordinal logistic regression model was used to identify the predictors of functional outcome and QOL at 6 months after stroke in ischemic and hemorrhagic stroke patients, respectively. Multivariate analysis with generalized linear mixed models was used for ordinal outcomes. The candidate predictors with a significance level of $P < 0.20$ identified in univariate ordinal logistic regression were entered into the generalized linear mixed models. The odds ratios of the significant predictors were generated from the analyses. P values less than 0.05 were considered statistically significant. All analyses were performed using SPSS version 21.0.

Results

Study population

Between August 2012 and October 2014, 8521 patients provided informed consent to participate in this follow-up study, with a participation rate of 78.3 % of the patients assessed for eligibility in the nine participating hospitals in Korea. Of 3002 stroke patients who completed the 6-month follow-up, 145 (4.8 %) were excluded due to missing information. Therefore, a total of 2857 first-time stroke patients were included in the final analysis (Fig. 1).

Patients characteristics

The basic characteristics of the patients are presented in Table 1. Of the subjects included in this study, 2289 patients (80.1 %) suffered from ischemic stroke, and 568 patients (19.9 %) experienced hemorrhagic stroke. The mean patient age was 64.3 years, and the ratio of males to females was 1.48:1.

Functional outcomes and quality of life at 6 months after stroke onset

The mean FIM scores (SD) at 6 months after stroke onset were 109.6 (27.8), 110.0 (27.3), and 108.0 (29.7) for total, ischemic, and hemorrhagic stroke patients, respectively. The proportions of patients with dependency at 6 months after stroke onset were 38.0 % ($n = 1086$), 37.9 % ($n = 868$), and 38.2 % ($n = 217$) in total, ischemic, and hemorrhagic stroke patients, respectively (Fig. 2).

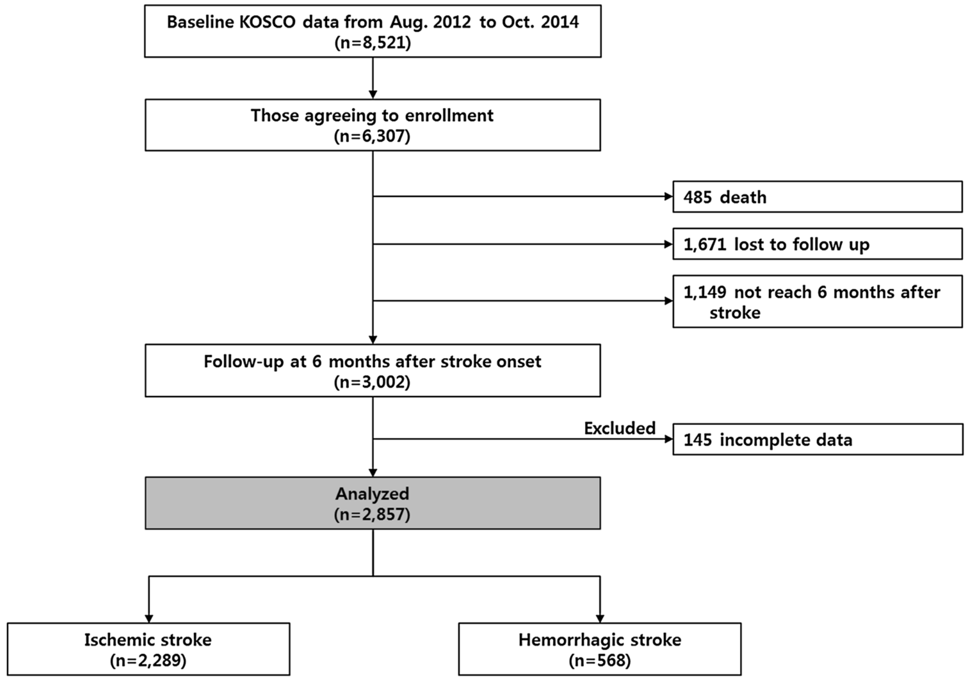

Fig. 1 Summary of the study population

The numbers of stroke patients who completed the survey for EQ-5D were 2271, 1839, and 432 in total, ischemic, and hemorrhagic stroke groups, respectively. The mean EQ-5D scores (SD) were 0.82488 (0.18644), 0.82411 (0.18660), and 0.82818 (9.18595) for total, ischemic, and hemorrhagic stroke patients, respectively. The proportions of patients with satisfaction at 6 months after stroke onset were 52.8 % ($n = 1199$), 52.1 % ($n = 958$), and 55.8 % ($n = 241$) in total, ischemic, and hemorrhagic stroke patients, respectively (Fig. 2).

There was a significant correlation between FIM and EQ-5D at 6 months after stroke onset in total, ischemic, and hemorrhagic stroke patients (Pearson correlation coefficient (PCC) = 0.776 ($P < 0.001$), PCC = 0.790 ($P < 0.001$), and PCC = 0.587 ($P < 0.001$), respectively).

Predictors of functional outcome at 6 months after stroke onset

The results from univariate and multivariate logistic regression analysis are summarized in Table 2. In the univariate analysis, many potential predictors demonstrated significant relationships with FIM at 6 months after stroke onset in patients with ischemic stroke ($P < 0.05$). The multivariate analysis showed that age, current alcohol consumption, initial NIHSS, hospitalization duration, and functional level at discharge were independent predictors of FIM at 6 months after onset for ischemic stroke ($P < 0.05$). Among these predictors, current alcohol consumption and all functional assessment scores at discharge were positive predictors of functional independency in the chronic phase in ischemic stroke patients.

In patients with hemorrhagic stroke, the univariate analysis showed that many potential predictors showed significant relationships with FIM at 6 months after stroke onset ($P < 0.05$). The multivariate analysis demonstrated that FMA, FAC, and short K-FAST score at discharge were positive predictors of FIM at 6 months after hemorrhagic stroke ($P < 0.05$).

Predictors of quality of life at 6 months after stroke onset

The results from univariate and multivariate logistic regression analysis are summarized in Table 3. The univariate analysis showed that many potential predictors had a significant relationship with EQ-5D at 6 months after ischemic stroke ($P < 0.05$). The multivariate analysis showed that age, sex, education duration, initial NIHSS, hospitalization duration, FMA, FAC, and short K-FAST at discharge were independent predictors of EQ-5D at 6 months after ischemic stroke ($P < 0.05$). Among these predictors, years of education, motor, ambulatory function, and language function at discharge were positive predictors for QOL in the chronic phase in ischemic stroke patients.

In patients with hemorrhagic stroke, the univariate analysis showed that many potential predictors had a significant relationship with EQ-5D at 6 months after stroke onset ($P < 0.05$). The multivariate analysis showed that

Table 1 Distribution of general and clinical patient characteristics

Variables	Mean ± SD or percentage		
	Total (*n* = 2857)	Ischemic (*n* = 2289)	Hemorrhagic (*n* = 568)
Age, years	64.3 ± 12.8	65.7 ± 12.4	58.5 ± 12.8
Sex, male	59.6	62.2	49.3
Body mass index	23.7 ± 3.3	23.8 ± 3.3	23.5 ± 3.2
Smoking, current	27.6	28.5	23.8
Alcohol, current	39.3	38.3	43.3
Education, years	9.6 ± 4.8	9.4 ± 4.8	10.4 ± 4.6
Medical history			
Hypertension, yes	54.4	56.7	45.1
Diabetes mellitus, yes	23.2	26.4	10.6
Coronary heart disease, yes	6.4	7.5	2.3
Atrial fibrillation, yes	10.3	12.3	2.1
Hyperlipidemia, yes	12.5	14.4	5.1
CCAS	5.4 ± 1.7	5.5 ± 1.7	4.9 ± 1.6
Premorbid mRS, score	0.9 ± 1.5	0.8 ± 1.4	1.1 ± 1.7
NIHSS initial, score	–	3.5 ± 4.8	–
GCS initial, score	–	–	12.5 ± 3.8
Time to arrival at hospital, h	22.2 ± 36.7	23.7 ± 37.8	16.1 ± 30.8
Initial aggravation, yes	5.3	5.9	2.6
Complication during hospitalization			
Thromboembolic disease, yes	1.5	1.7	0.9
Pneumonia, yes	2.8	2.2	5.3
Ventilatory insufficiency, yes	0.6	0.5	1.2
Urinary tract infection, yes	3.3	2.7	5.8
Bladder dysfunction, yes	6.7	5.6	10.9
Bowel dysfunction, yes	9.8	8.9	13.4
Insomnia, yes	6.8	6.4	8.1
Number of complications (0–7)	0.3 ± 0.7	0.3 ± 0.7	0.5 ± 0.8
Duration of hospitalization, days	18.2 ± 25.0	14.8 ± 22.6	16.1 ± 30.8
Functional level at discharge			
K-MMSE	21.9 ± 8.9	21.9 ± 8.9	21.6 ± 9.2
Fugl-Meyer assessment	79.8 ± 31.6	79.5 ± 31.7	81.1 ± 31.3
Functional ambulatory category	3.3 ± 1.7	3.3 ± 1.9	3.0 ± 1.8
AHSA-NOMS	6.3 ± 1.5	6.4 ± 1.4	6.3 ± 1.6
Short K-FAST	13.4 ± 6.1	13.4 ± 6.0	13.1 ± 6.5
Classification of ischemic stroke subtype			
Large artery atherosclerosis	–	47.9	
Small vessel occlusion	–	22.0	
Cardioembolism	–	12.7	
Other determined	–	5.8	
Undetermined	–	11.6	
Classification of hemorrhagic stroke subtype[a]			
Intracerebral hemorrhage	–	–	59.9
Subarachnoid hemorrhage	–	–	50.5
Intraventricular hemorrhage	–	–	16.2
Subdural hematoma	–	–	2.8
Epidural hematoma	–	–	0.4

CCAS combined condition- and age-related score, *NIHSS* National Institutes of Health Stroke Scale, *mRS* modified Rankin scale, *REH* Rehabilitation Department, *K-MMSE* Korean Mini-Mental State Examination, *AHSA-NOMS*, the American Speech-Language-Hearing Association National Outcome Measurement System Swallowing Scale, *Short K-FAST* Short Korean Version of the Frenchay Aphasia Screening test

[a] Multiresponse analysis

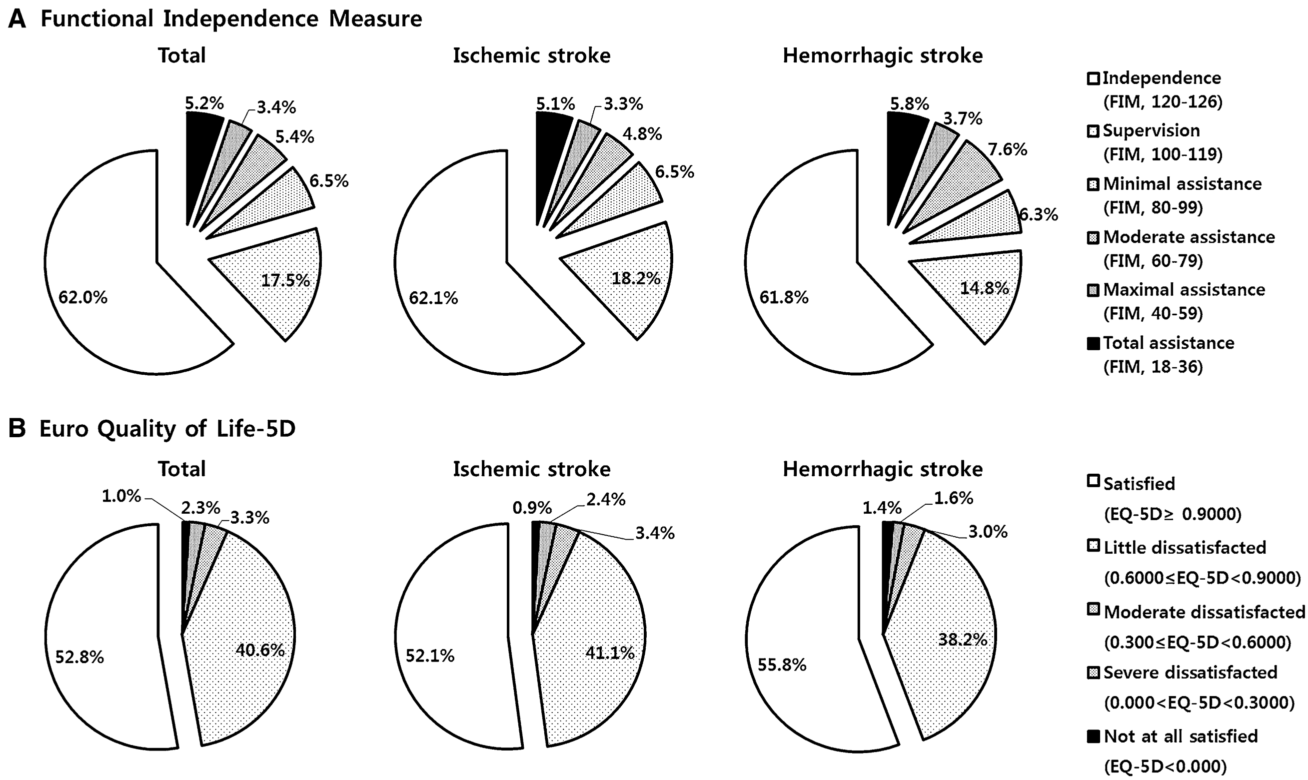

Fig. 2 Functional level and quality of life at 6 months after stroke onset. *FIM* functional independence measurement, *EQ-5D* Euro Quality of Life-5D

FMA score at discharge was a positive predictor of EQ-5D at 6 months after hemorrhagic stroke ($P < 0.05$).

Discussion

This study revealed that a large proportion of stroke survivors were not independent in their daily living activities and suffered from low QOL even 6 months after stroke onset. Analysis demonstrated that age, initial stroke severity, duration of hospitalization, and functional level at discharge were independent predictors of functional level in the chronic phase in stroke survivors. In addition, previous alcohol consumption and experience with pneumonia during hospitalization were independent predictors of functional level in survivors after ischemic and hemorrhagic stroke, respectively. For QOL, age, duration of hospitalization, and motor function at discharge were independent predictors in the chronic phase in stroke survivors.

Predictors that have a significant impact on the chronic phases of stroke are important in terms of continued functional recovery. Many variables are known to predict functional outcome in stroke patients [5]. However, the analysis of predictors should be considered in order to eliminate confounding factors that often interact with each other. In this prospective cohort study, a generalized linear mixed model was used to explore the influence of predictors on functional outcome and QOL in first-ever stroke survivors. We excluded cases of death in the analysis of factors that influenced functional outcome and QOL. The significant independent predictors in this study were separated into non-modifiable and modifiable factors for both ischemic and hemorrhagic stroke patients. The common independent non-modifiable predictors for functional outcome in both ischemic and hemorrhagic stroke were age and initial stroke severity. These results were consistent with those of previous large prospective cohort studies [30, 31] that reported age and stroke severity in the early stroke phase to be strongly associated with ADL outcome in the chronic phase. The duration of hospitalization was a significant independent negative modifiable predictor of functional level in the chronic phase. Hospitalization duration may represent the severity of medical complications after stroke onset [32]. These could mean that prevention and reduction of medical complications after stroke should be considered the goals for hospitalized stroke patients. In addition, the motor, ambulatory, and language functions at discharge were significant independent modifiable predictors of functional level in the chronic phase for

Table 2 Regression analysis for potential predictors of functional outcome at 6 months after stroke

	Ischemic stroke (n = 2289)						Hemorrhagic stroke (n = 568)					
	Univariate analysis			Multivariate analysis			Univariate analysis			Multivariate analysis		
	P	OR	95 % CI	*P*	OR	95 % CI	*P*	OR	95 % CI	*P*	OR	95 % CI
Age	<0.001*	0.937	0.929–0.945	<0.001*	0.945	0.935–0.956	<0.001*	0.937	0.924–0.951	<0.001*	0.948	0.926–0.971
Sex, female	<0.001*	0.507	0.429–0.599	0.367	0.892	0.695–1.144	0.030*	0.694	0.500–0.965	0.860	0.956	0.579–1.578
Body mass index	<0.001*	1.097	1.068–1.126	0.669	1.007	0.976–1.039	0.088	1.416	0.950–2.112	0.780	1.009	0.946–1.077
Smoking, current	<0.001*	1.424	1.182–1.715	0.220	1.180	0.906–1.639	0.195	1.035	0.983–1.089	0.236	1.458	0.781–2.720
Alcohol, current	<0.001*	1.723	1.448–2.049	0.043*	1.285	1.008–1.536	0.001*	1.838	1.289–2621	0.162	1.437	0.864–2.391
Education years	<0.001*	1.116	1.095–1.137	0.295	1.013	0.989–1.039	<0.001*	1.129	1.087–1.172	0.620	0.986	0.935–1.041
Medical history												
Hypertension	0.001*	0.715	0.605–0.844	0.592	1.061	0.855–1.315	<0.001*	0.547	0.394–0.761	0.175	1.355	0.874–2.103
Diabetes mellitus	0.057	0.837	0.697–1.005	0.132	1.192	0.948–1.498	0.002*	0.448	0.273–0.735	0.401	1.324	0.688–2.547
Coronary heart disease	0.047*	0.739	0.548–0.996	0.200	1.273	0.880–1.840	0.405	1.675	0.497–5.641	–	–	–
Atrial fibrillation	0.001*	0.666	0.525–0.845	0.368	0.872	0.646–1.176	0.275	0.555	0.193–1.597	–	–	–
Hyperlipidemia	0.427	0.911	0.723–1.147	–	–	–	0.699	1.162	0.543–2.488	–	–	–
CCAS	<0.001*	0.902	0.860–0.946	0.658	0.987	0.929–1.048	<0.001*	0.780	0.705–0.863	0.938	0.994	0.864–1.145
Premorbid mRS	<0.001*	0.847	0.800–0.897	0.228	1.046	0.972–1.126	0.429	0.963	0.879–1.057	–	–	–
NIHSS initial	<0.001*	0.723	0.707–0.739	<0.001*	0.918	0.889–0.948	–	–	–	–	–	–
GCS initial	–	–	–	–	–	–	<0.001*	1.160	1.109–1.214	0.004*	1.088	1.027–1.153
Arrival time to hospital	0.226	0.902	0.860–0.860	–	–	–	0.019*	1.008	1.001–1.014	0.211	1.005	0.997–1.014
Neurologic aggravation	<0.001*	0.410	0.298–0.563	0.082	1.435	0.955–2.157	0.242	0.567	0.220–1.467	–	–	–
Complication during hospitalization												
Thromboembolic disease	0.556	1.220	0.629–2.369		–	—	0.782	1.296	0.206–8.143	–	–	–
Pneumonia	<0.001*	0.124	0.075–0.206	0.088	2.492	0.873–7.116	<0.001*	0.151	0.078–0.294	0.015*	4.204	1.320–13.391
Ventilatory insufficiency	<0.001*	0.091	0.031–0.262	0.149	3.313	0.652–16.832	0.004*	0.141	0.037–0.534	0.401	2.271	0.334–15.435
Urinary tract infection	<0.001*	0.145	0.092–0.229	0.229	1.859	0.677–5.103	0.196	0.646	0.333–1.252	0.507	1.551	0.424–5.677
Bladder dysfunction	<0.001*	0.210	0.152–0.290	0.275	1.743	0.642–4.730	<0.001*	0.346	0.213–0.561	0.621	1.291	0.469–3.550
Bowel dysfunction	<0.001*	0.412	0.316–0.538	0.365	1.524	0.613–3.789	0.104	0.682	0.430–1.082	0.550	1.497	0.398–5.631
Insomnia	0.057	0.733	0.532–1.010	0.093	2.334	0.867–6.281	0.402	1.307	0.699–2.444	–	–	
N. of complications	<0.001*	0.536	0.479–0.601	0.270	1.608	0.692–3.737	<0.001*	0.667	0.552–0.805	0.143	1.894	0.806–4.453
Duration of hospitalization	<0.001*	0.965	0.960–0.969	<0.001*	0.979	0.973–0.984	<0.001*	0.973	0.967–0.979	<0.001*	0.979	0.972–0.986
Functional level at discharge												
K-MMSE	<0.001*	1.151	1.139–1.164	<0.001*	1.054	1.038–1.070	<0.001*	1.160	1.136–1.184	0.076	1.037	0.996–1.079
FMA	<0.001*	1.042	1.039–1.045	<0.001*	1.019	1.015–1.023	<0.001*	1.047	1.040–1.054	<0.001*	1.023	1.014–1.032

Table 2 continued

	Ischemic stroke ($n = 2289$)						Hemorrhagic stroke ($n = 568$)					
	Univariate analysis			Multivariate analysis			Univariate analysis			Multivariate analysis		
	P	OR	95 % CI	*P*	OR	95 % CI	*P*	OR	95 % CI	*P*	OR	95 % CI
FAC	<0.001*	2.490	2.343–2.648	<0.001*	1.420	1.306–1.544	<0.001*	2.695	2.376–3.056	<0.001*	1.516	1.267–1.813
AHSA-NOMS	<0.001*	1.747	1.665–1.833	<0.001*	1.151	1.070–1.238	<0.001*	1.946	1.741–2.175	0.139	1.115	0.965–1.289
Short K-FAST	<0.001*	1.252	1.231–1.272	<0.001*	1.055	1.028–1.082	<0.001*	1.254	1.216–1.294	0.046*	1.062	1.001–1.127

CCAS combined condition- and age-related score, *NIHSS* National Institutes of Health Stroke Scale, *mRS* modified Rankin scale, *K-MMSE* Korean Mini-Mental State Examination, *FMA* Fugl-Meyer assessment, *FAC* functional ambulatory category, *AHSA-NOMS* the American Speech-Language-Hearing Association National Outcome Measurement System Swallowing Scale, *Short K-FAST* Short Korean Version of the Frenchay Aphasia Screening test

* $P < 0.05$

both ischemic and hemorrhagic stroke patients. Previous studies have reported that motor weakness, gait function, and language function in acute and subacute phases are related to functional outcome in stroke patients in the chronic phase [4, 33–35]. However, a novel finding in this study was that each functional domain score at discharge was independently related with functional outcome in the chronic phase in ischemic and hemorrhagic stroke survivors. In addition, the strongest predictor in each functional domain in the chronic phase in stroke patients was the functional level at discharge [4, 36]. These results indicate that functional independency in stroke patients relies on recovery from impairment in all functional domains. Therefore, assessment of each functional domain and reducing impairments during hospitalization are critical for improving functional outcomes in the chronic phase in stroke survivors.

In this interim analysis of KOSCO, pneumonia during the initial hospitalization was a significant predictor of functional outcomes in both ischemic and hemorrhagic stroke patients at 6 months after onset. However, it was a significant independent predictor of functional outcomes only in hemorrhagic stroke patients. In previous studies, the infection rate during the initial hospitalization (ranging from 5 to 65 %) was higher than the infection rate measured at 6 months after stroke in this study [37]. Infection during the initial hospitalization is a major cause of mortality after stroke, so the infection rate may be lower when measured 6 months after stroke because infected patients who died within 6 months are not included [32]. In addition, the lower incidence of pneumonia in ischemic stroke compared with hemorrhagic stroke could contribute to the differences observed in this study. Because of these factors, our study does not address the influence of infection during the initial hospitalization on recovery of function in ischemic stroke patients. However, this study addressed other strong predictors of functional level and quality of life in stroke patients.

In this study, alcohol consumption before stroke onset was an independent predictor of functional level in survivors of ischemic stroke. Long-term effects of alcohol consumption can negatively affect the brain [38]; however, not all effects of alcohol consumption are harmful. Instead, light to moderate drinking is known to be beneficial for cardiovascular health [39]. In the present study, alcohol consumption was recorded during medical chart review, and the amount of alcohol or duration of continuous alcohol consumption was not assessed. Therefore, further study is needed to determine characteristics of alcohol consumption and effects on functional outcome in stroke patients.

QOL showed a significant correlation with functional level in the chronic phase in stroke survivors. This result is

Table 3 Regression analysis for potential predictors of quality of life at 6 months after stroke

	Ischemic stroke ($n = 1839$)						Hemorrhagic stroke ($n = 432$)					
	Univariate analysis			Multivariate analysis			Univariate analysis			Multivariate analysis		
	P	OR	95 % CI	*P*	OR	95 % CI	*P*	OR	95 % CI	*P*	OR	95 % CI
Age	<0.001*	0.949	0.943–0.956	<0.001*	0.966	0.955–0.977	<0.001*	0.950	0.938–0.962	0.002*	0.960	0.935–0.985
Sex, female	<0.001*	0.566	0.484–0.662	0.046*	0.774	0.602–0.995	0.036*	0.722	0.533–0.979	0.401	0.824	0.524–1.295
Body mass index	0.014*	1.282	1.052–1.563	0.309	1.017	0.985–1.050	0.769	0.938	0.611–1.440	–	–	–
Smoking, current	<0.001*	1.072	1.047–1.098	0.119	1.226	0.949–1.583	0.355	1.023	0.975–1.072	–	–	–
Alcohol, current	<0.001*	1.593	1.322–1.919	0.228	1.155	0.914–1.461	0.642	1.095	0.747–1.603	–	–	–
Education years	<0.001*	1.104	1.082–1.126	0.004*	1.038	1.012–1.064	<0.001*	1.100	1.053–1.149	0.092	1.049	0.992–1.109
Medical history												
Hypertension	<0.001*	0.757	0.650–0.883	0.741	1.037	0.837–1.285	<0.001*	0.565	0.416–0.768	0.027*	1.653	1.060–2.577
Diabetes mellitus	0.056	0.847	0.714–1.005	0.883	1.018	0.804–1.288	0.026*	0.575	0.353–0.937	0.946	0.974	0.455–2.087
Coronary heart disease	0.707	0.947	0.711–1.261	–	–	–	0.277	1.805	0.622–5.237	–	–	–
Atrial fibrillation	0.012*	0.747	0.594–0.938	0.217	0.818	0.595–1.125	0.201	0.506	0.178–1.437	–	–	–
Hyperlipidemia	0.685	0.956	0.772–1.186	–	–	–	0.439	0.764	0.386–1.510	–	–	–
CCAS	<0.001*	0.921	0.882–0.963	0.110	0.947	0.885–1.013	<0.001*	0.806	0.732–0.887	0.415	1.084	0.893–1.314
Premorbid mRS	<0.001*	0.864	0.818–0.913	0.202	1.054	0.972–1.142	0.857	1.008	0.924–1.099	–	–	–
NIHSS initial	<0.001*	0.800	0.784–0.816	<0.001*	0.896	0.858–0.936	–	–	–	–	–	–
GCS initial	–	–	–	–	–	–	0.039*	1.058	1.003–1.117	0.412	1.026	0.964–1.093
Arrival time to hospital	0.154	1.001	0.999–1.004	0.083	0.998	0.995–1.000	0.083	1.005	0.999–1.010	0.757	1.001	0.994–1.008
Neurologic aggravation	<0.001*	0.526	0.383–0.721	0.289	1.268	0.818–1.964	0.022*	0.329	0.127–0.853	0.023*	5.331	1.253–22.681
Complication during hospitalization												
Thromboembolic disease	0.627	0.865	0.481–1.556	–	–	–	0.692	0.724	0.146–3.597	–	–	–
Pneumonia	<0.001*	0.246	0.146–0.414	0.920	0.942	0.295–3.007	<0.001*	0.269	0.135–0.539	0.023*	4.391	1.232–15.654
Ventilatory insufficiency	0.034*	0.307	0.103–0.913	0.493	0.491	0.064–3.763	0.069	0.272	0.067–1.108	0.581	0.457	0.028–7.346
Urinary tract infection	<0.001*	0.231	0.144–0.373	0.855	0.903	0.303–2.694	0.862	0.944	0.485–1.801	–	–	–
Bladder dysfunction	<0.001*	0.270	0.194–0.376	0.875	1.082	0.407–2.873	<0.001*	0.405	0.249–0.658	0.557	1.345	0.501–3.613
Bowel dysfunction	<0.001*	0.449	0.345–0.584	0.441	1.422	0.581–3.482	0.006*	0.541	0.348–0.841	0.515	1.549	0.414–5.795
Insomnia	0.158	0.802	0.591–1.089	0.582	0.769	0.301–1.960	0.314	0.754	0.435–1.307	–	–	–
N. of complications	<0.001*	0.607	0.542–0.680	0.940	0.971	0.448–2.104	<0.001*	0.670	0.556–0.807	0.337	1.382	0.713–2.679
Duration of hospitalization	<0.001*	0.976	0.972–0.981	<0.001*	0.977	0.970–0.984	<0.001*	0.982	0.976–0.987	<0.001*	0.982	0.973–0.991
Functional level at discharge												
K-MMSE	<0.001*	1.130	1.119–1.142	0.713	0.996	0.975–1.017	<0.001*	1.121	1.099–1.144	0.837	1.005	0.956–1.057
FMA	<0.001*	1.033	1.030–1.036	<0.001*	1.011	1.006–1.017	<0.001*	1.032	1.027–1.038	<0.001*	1.028	1.016–1.041
FAC	<0.001*	1.831	1.739–1.928	<0.001*	1.397	1.274–1.532	<0.001*	1.739	1.582–1.912	0.063	1.205	0.990–1.467

Table 3 continued

	Ischemic stroke (n = 1839)						Hemorrhagic stroke (n = 432)					
	Univariate analysis			Multivariate analysis			Univariate analysis			Multivariate analysis		
	P	OR	95 % CI	*P*	OR	95 % CI	*P*	OR	95 % CI	*P*	OR	95 % CI
AHSA-NOMS	<0.001*	1.671	1.571–1.779	0.733	0.982	0.888–1.088	<0.001*	1.730	1.532–1.953	0.421	1.102	0.869–1.398
Short K-FAST	<0.001*	1.147	1.124–1.169	0.013*	1.039	1.008–1.072	<0.001*	1.120	1.079–1.162	0.600	0.982	0.916–1.052

CCAS combined condition- and age-related score, *NIHSS* National Institutes of Health Stroke Scale, *mRS* modified Rankin scale, *REH* Rehabilitation Department, *K-MMSE* Korean Mini-Mental State Examination, *FMA* Fugl-Meyer Assessment, *FAC* functional ambulatory category, *AHSA-NOMS* the American Speech-Language-Hearing Association National Outcome Measurement System Swallowing Scale, *Short K-FAST* Short Korean Version of the Frenchay Aphasia Screening test

* $P < 0.05$

consistent with findings from previous reports [40, 41]. However, the predictors of QOL were different from those of functional level in chronic phase stroke survivors, suggesting that QOL in stroke patients is influenced by factors other than functional level [42]. In particular, motor function at discharge was an independent predictor of QOL in the chronic phase in patients with both ischemic and hemorrhagic stroke. However, this result might be due to the statistical advantage of FMA with a wider range than that of the other functional domain assessments. Another possible explanation is a bias in the relatively higher cognitive and language functions in the study population with regard to the EQ-5D, which demands appropriate cognition and language function. Therefore, the direct relationship between motor function at discharge and QOL in the chronic phase requires further investigation. In spite of this limitation, the motor function measured by FMA at discharge could be used as a valuable factor to predict QOL in chronic stroke patients. Education duration was an independent predictor of QOL in ischemic stroke survivors and showed a tendency with QOL in hemorrhagic stroke survivors as well; however, it was not an independent predictor of functional outcome. Education level is associated with socioeconomic status [43], and the socioeconomic status itself is significantly correlated with QOL [44]. However, the relationship between socioeconomic status and functional outcome is unclear in stroke patients [43], possibly because functional outcome can be influenced by multiple factors in stroke patients [8–10]. This study used multivariate analysis to control for confounding factors to confirm or refute these associations. These results were consistent with those from previous studies that reported no association or a weak association between socioeconomic status and stroke outcome [45, 46]. Therefore, socioeconomic status was an independent predictor of QOL but not functional outcome in chronic stroke survivors.

The study had some strengths and limitations. The primary strength of this study was the fact that it was a large multicenter study of functional outcome and QOL in acute and chronic phase stroke patients who suffered a first-ever stroke. The face-to-face interactions with patients enabled accurate evaluation. In addition, we analyzed the data according to ischemic and hemorrhagic stroke. However, one of the limitations of this study was a relatively high proportion of missing information (4.8 %). We included only completed data for most variables that might influence functional level or QOL in chronic stroke patients. Although it could lead to a bias in the results, this approach resulted in a more accurate analysis. The face-to-face survey with EQ-5D to assess QOL required appropriate cognition and language function. Based on this requirement, 79.5 % of stroke survivors were able to perform the EQ-5D, and this was another limitation of the present study. In addition, assessment of depression and anxiety at discharge from the initial hospitalization was not included in this study. Emotional status is an important predictor of functional level and quality of life at the chronic stage in stroke patients [16]. Because emotional assessments are required for proper cognitive function, many subjects could not complete a questionnaire of emotional status at discharge from the initial hospitalization. This is another limitation in this study. Despite many limitations, this study is a useful contribution to the literature on identifying factors for improved stroke outcomes and expands the evidence base to allow for comparisons between different nationalities.

In conclusion, this study found that functional level and QOL in chronic stroke patients partially share the acute phase predictors. Motor function at discharge from the first admission is important predictor that is modifiable by qualified in-hospital care. Proper treatment to achieve maximal functional gain at discharge may be an important factor in improving functional independency and QOL in chronic stage stroke survivors. The present study provides useful information for establishing comprehensive and systematic care for stroke patients.

Acknowledgments This work was supported by the Research Program funded by the Korea Centers for Disease Control and Prevention (2013-2013E3301702) and by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIP) (NRF-2014R1A2A1A01005128).

Compliance with ethical standards

Conflicts of interest The authors report no conflicts of interest.

References

1. Mukherjee D, Patil CG (2011) Epidemiology and the global burden of stroke. World Neurosurg 76:S85–S90
2. Barker-Collo S, Feigin VL, Parag V, Lawes CM, Senior H (2010) Auckland stroke outcomes study. Part 2: cognition and functional outcomes 5 years poststroke. Neurology 75:1608–1616
3. Langhorne P, Bernhardt J, Kwakkel G (2011) Stroke rehabilitation. Lancet 377:1693–1702
4. Veerbeek JM, Kwakkel G, van Wegen EE, Ket JC, Heymans MW (2011) Early prediction of outcome of activities of daily living after stroke: a systematic review. Stroke 42:1482–1488
5. Kwakkel G, Kollen BJ (2013) Predicting activities after stroke: what is clinically relevant? Int J Stroke 8:25–32
6. Weimar C, Konig IR, Kraywinkel K, Ziegler A, Diener HC, German Stroke Study C (2004) Age and National Institutes of Health Stroke Scale Score within 6 hours after onset are accurate predictors of outcome after cerebral ischemia: development and external validation of prognostic models. Stroke 35:158–162
7. Feigin VL, Barker-Collo S, Parag V, Senior H, Lawes CM, Ratnasabapathy Y, Glen E, Group AS (2010) Auckland stroke outcomes study. Part 1: gender, stroke types, ethnicity, and functional outcomes 5 years poststroke. Neurology 75:1597–1607
8. Bernhardt J, Dewey H, Thrift A, Collier J, Donnan G (2008) A very early rehabilitation trial for stroke (AVERT): phase II safety and feasibility. Stroke 39:390–396
9. Slot KB, Berge E, Dorman P, Lewis S, Dennis M, Sandercock P, Oxfordshire Community Stroke Project tIST, Lothian Stroke R (2008) Impact of functional status at 6 months on long term survival in patients with ischaemic stroke: prospective cohort studies. BMJ 336:376–379
10. Toschke AM, Tilling K, Cox AM, Rudd AG, Heuschmann PU, Wolfe CD (2010) Patient-specific recovery patterns over time measured by dependence in activities of daily living after stroke and post-stroke care: the South London Stroke Register (SLSR). Eur J Neurol 17:219–225
11. Grube MM, Koennecke HC, Walter G, Thummler J, Meisel A, Wellwood I, Heuschmann PU, Berlin Stroke R (2012) Association between socioeconomic status and functional impairment 3 months after ischemic stroke: the Berlin Stroke Register. Stroke 43:3325–3330
12. Putman K, De Wit L, Schoonacker M, Baert I, Beyens H, Brinkmann N, Dejaeger E, De Meyer AM, De Weerdt W, Feys H, Jenni W, Kaske C, Leys M, Lincoln N, Schuback B, Schupp W, Smith B, Louckx F (2007) Effect of socioeconomic status on functional and motor recovery after stroke: a European multicentre study. J Neurol Neurosurg Psychiatry 78:593–599
13. Jakovljevic D, Sarti C, Sivenius J, Torppa J, Mahonen M, Immonen-Raiha P, Kaarsalo E, Alhainen K, Kuulasmaa K, Tuomilehto J, Puska P, Salomaa V (2001) Socioeconomic status and ischemic stroke: the FINMONICA stroke register. Stroke 32:1492–1498
14. van den Bos GA, Smits JP, Westert GP, van Straten A (2002) Socioeconomic variations in the course of stroke: unequal health outcomes, equal care? J Epidemiol Community Health 56:943–948
15. Kerr GD, Higgins P, Walters M, Ghosh SK, Wright F, Langhorne P, Stott DJ (2011) Socioeconomic status and transient ischaemic attack/stroke: a prospective observational study. Cerebrovasc Dis 31:130–137
16. Guajardo VD, Terroni L, Sobreiro Mde F, Zerbini MI, Tinone G, Scaff M, Iosifescu DV, de Lucia MC, Fraguas R (2015) The influence of depressive symptoms on quality of life after stroke: a prospective study. J Stroke Cerebrovasc Dis 24:201–209
17. Chang WH, Sohn MK, Lee J, Kim DY, Lee SG, Shin YI, Oh GJ, Lee YS, Joo MC, Han EY, Kim YH (2015) Korean Stroke Cohort for functioning and rehabilitation (KOSCO): study rationale and protocol of a multi-centre prospective cohort study. BMC Neurol 15:42
18. Dodds TA, Martin DP, Stolov WC, Deyo RA (1993) A validation of the functional independence measurement and its performance among rehabilitation inpatients. Arch Phys Med Rehabil 74:531–536
19. Greiner W, Claes C, Busschbach JJ, von der Schulenburg JM (2005) Validating the EQ-5D with time trade off for the German population. Eur J Health Econ 6:124–130
20. Kang EHS (2006) A valuation of health status using EQ-5D. Korean J Health Econ Policy 12:19–43
21. Oh MS, Yu KH, Lee JH, Jung S, Ko IS, Shin JH, Cho SJ, Choi HC, Kim HH, Lee BC (2012) Validity and reliability of a Korean Version of the National Institutes of Health Stroke Scale. J Clin Neurol 8:177–183
22. Teasdale G, Knill-Jones R, van der Sande J (1978) Observer variability in assessing impaired consciousness and coma. J Neurol Neurosurg Psychiatry 41:603–610
23. Kang Y, Na DL, Hahn S (1997) A validity study on the Korean Mini-Mental State Examination (K-MMSE) in dementia patients. J Korean Neurol Assoc 15:300–308

24. Fugl-Meyer AR, Jaasko L, Leyman I, Olsson S, Steglind S (1975) The post-stroke hemiplegic patient. 1. A method for evaluation of physical performance. Scand J Rehabil Med 7:13–31
25. Holden MK, Gill KM, Magliozzi MR, Nathan J, Piehl-Baker L (1984) Clinical gait assessment in the neurologically impaired. Reliability and meaningfulness. Phys Ther 64:35–40
26. Wesling M, Brady S, Jensen M, Nickell M, Statkus D, Escobar N (2003) Dysphagia outcomes in patients with brain tumors undergoing inpatient rehabilitation. Dysphagia 18:203–210
27. Pyun SB, Hwang YM, Ha JW, Yi H, Park KW, Nam K (2009) Standardization of Korean Version of Frenchay Aphasia Screening test in normal adults. J Korean Acad Rehabil Med 33:436–440
28. Bernardini J, Callen S, Fried L, Piraino B (2004) Inter-rater reliability and annual rescoring of the Charlson comorbidity index. Adv Perit Dial 20:125–127
29. Burn JP (1992) Reliability of the modified Rankin Scale. Stroke 23:438
30. Kwakkel G, Veerbeek JM, Harmeling-van der Wel BC, van Wegen E, Kollen BJ, Early Prediction of functional Outcome after Stroke I (2011) Diagnostic accuracy of the Barthel Index for measuring activities of daily living outcome after ischemic hemispheric stroke: does early poststroke timing of assessment matter? Stroke 42:342–346
31. Heuschmann PU, Wiedmann S, Wellwood I, Rudd A, Di Carlo A, Bejot Y, Ryglewicz D, Rastenyte D, Wolfe CD, European Registers of S (2011) Three-month stroke outcome: the European Registers of Stroke (EROS) investigators. Neurology 76:159–165
32. Ingeman A, Andersen G, Hundborg HH, Svendsen ML, Johnsen SP (2011) In-hospital medical complications, length of stay, and mortality among stroke unit patients. Stroke 42:3214–3218
33. Parker VM, Wade DT, Langton Hewer R (1986) Loss of arm function after stroke: measurement, frequency, and recovery. Int Rehabil Med 8:69–73
34. Meijer R, van Limbeek J, Peusens G, Rulkens M, Dankoor K, Vermeulen M, de Haan RJ (2005) The stroke unit discharge guideline, a prognostic framework for the discharge outcome from the hospital stroke unit. A prospective cohort study. Clin Rehabil 19:770–778
35. Kim BR, Han EY, Joo SJ, Kim SY, Yoon HM (2014) Cardiovascular fitness as a predictor of functional recovery in subacute stroke patients. Disabil Rehabil 36:227–231
36. Gialanella B, Ferlucci C (2010) Functional outcome after stroke in patients with aphasia and neglect: assessment by the motor and cognitive functional independence measure instrument. Cerebrovasc Dis 30:440–447
37. Westendorp WF, Nederkoorn PJ, Vermeij JD, Dijkgraaf MG, van de Beek D (2011) Post-stroke infection: a systematic review and meta-analysis. BMC Neurol 11:110
38. Bartley PC, Rezvani AH (2012) Alcohol and cognition—consideration of age of initiation, usage patterns and gender: a brief review. Curr Drug Abuse Rev 5:87–97
39. Holmes MV, Dale CE, Zuccolo L, Silverwood RJ, Guo Y, Ye Z, Prieto-Merino D, Dehghan A, Trompet S, Wong A, Cavadino A, Drogan D, Padmanabhan S, Li S, Yesupriya A, Leusink M, Sundstrom J, Hubacek JA, Pikhart H, Swerdlow DI, Panayiotou AG, Borinskaya SA, Finan C, Shah S, Kuchenbaecker KB, Shah T, Engmann J, Folkersen L, Eriksson P, Ricceri F, Melander O, Sacerdote C, Gamble DM, Rayaprolu S, Ross OA, McLachlan S, Vikhireva O, Sluijs I, Scott RA, Adamkova V, Flicker L, Bockxmeer FM, Power C, Marques-Vidal P, Meade T, Marmot MG, Ferro JM, Paulos-Pinheiro S, Humphries SE, Talmud PJ, Mateo Leach I, Verweij N, Linneberg A, Skaaby T, Doevendans PA, Cramer MJ, van der Harst P, Klungel OH, Dowling NF, Dominiczak AF, Kumari M, Nicolaides AN, Weikert C, Boeing H, Ebrahim S, Gaunt TR, Price JF, Lannfelt L, Peasey A, Kubinova R, Pajak A, Malyutina S, Voevoda MI, Tamosiunas A, Maitland-van der Zee AH, Norman PE, Hankey GJ, Bergmann MM, Hofman A, Franco OH, Cooper J, Palmen J, Spiering W, de Jong PA, Kuh D, Hardy R, Uitterlinden AG, Ikram MA, Ford I, Hypponen E, Almeida OP, Wareham NJ, Khaw KT, Hamsten A, Husemoen LL, Tjonneland A, Tolstrup JS, Rimm E, Beulens JW, Verschuren WM, Onland-Moret NC, Hofker MH, Wannamethee SG, Whincup PH, Morris R, Vicente AM, Watkins H, Farrall M, Jukema JW, Meschia J, Cupples LA, Sharp SJ, Fornage M, Kooperberg C, LaCroix AZ, Dai JY, Lanktree MB, Siscovick DS, Jorgenson E, Spring B, Coresh J, Li YR, Buxbaum SG, Schreiner PJ, Ellison RC, Tsai MY, Patel SR, Redline S, Johnson AD, Hoogeveen RC, Hakonarson H, Rotter JI, Boerwinkle E, de Bakker PI, Kivimaki M, Asselbergs FW, Sattar N, Lawlor DA, Whittaker J, Davey Smith G, Mukamal K, Psaty BM, Wilson JG, Lange LA, Hamidovic A, Hingorani AD, Nordestgaard BG, Bobak M, Leon DA, Langenberg C, Palmer TM, Reiner AP, Keating BJ, Dudbridge F, Casas JP, InterAct C (2014) Association between alcohol and cardiovascular disease: Mendelian randomisation analysis based on individual participant data. BMJ 349:g4164
40. Hartman-Maeir A, Soroker N, Ring H, Avni N, Katz N (2007) Activities, participation and satisfaction 1-year post stroke. Disabil Rehabil 29:559–566
41. Baumann M, Le Bihan E, Chau K, Chau N (2014) Associations between quality of life and socioeconomic factors, functional impairments and dissatisfaction with received information and home-care services among survivors living at home 2 years after stroke onset. BMC Neurol 14:92
42. van de Port IG, Kwakkel G, Schepers VP, Heinemans CT, Lindeman E (2007) Is fatigue an independent factor associated with activities of daily living, instrumental activities of daily living and health-related quality of life in chronic stroke? Cerebrovasc Dis 23:40–45
43. Cox AM, McKevitt C, Rudd AG, Wolfe CD (2006) Socioeconomic status and stroke. Lancet Neurol 5:181–188
44. Hudson CG (2005) Socioeconomic status and mental illness: tests of the social causation and selection hypotheses. Am J Orthopsychiatry 75:3–18
45. Aslanyan S, Weir CJ, Lees KR, Reid JL, McInnes GT (2003) Effect of area-based deprivation on the severity, subtype, and outcome of ischemic stroke. Stroke 34:2623–2628
46. Weir NU, Gunkel A, McDowall M, Dennis MS (2005) Study of the relationship between social deprivation and outcome after stroke. Stroke 36:815–819